

Ishpreet Nagi

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EDUCATION

McMaster University

Bachelor of Applied Science, Computer Science

Hamilton, ON

Sept 2021 — May 2026

- Cumulative GPA: Summa Cum Laude | Dean's Honor List
- Relevant Coursework: Data Structures, Algorithms, Databases, Operating Systems, Distributed Systems, Concurrency, Human-Computer Interfaces, Applications of Machine Learning, Discrete Mathematics

WORK EXPERIENCE

Software Engineer and Research Coordinator

June 2024 — July 2025

Healthcare Systems Research & Analysis

Waterloo, ON

- Shipped end-to-end customer-facing features across the full stack, owning technical design, implementation, code review, and production rollout for a data classification product used across multiple healthcare regions
- Designed and built scalable REST APIs and a distributed backend pipeline in Python with PostgreSQL and Pandas, cutting manual analyst workflows and **automating 90% of downstream deployment**
- Integrated an ML classification service behind a documented REST interface, delivering **98% accuracy** while remaining reliable under production load
- Led a team of 4 engineers across 8 stakeholders, driving architecture decisions, enforcing test coverage and code review standards, and shipping every milestone on time

Machine Learning Analyst and Research Assistant

May 2023 — May 2025

McMaster University - Department of Computing and Software

Hamilton, ON

- Built and maintained an automated evaluation service under Dr. Douglas Down, orchestrating Optuna-based hyperparameter search and reproducible experiment tracking to **improve model effectiveness by 20%**
- Refactored a legacy research codebase into a modular, well-tested Python package with CI checks, **reducing iteration time by 40%** and enabling other researchers to extend it independently

PROJECTS

Kollec App (kollec.app)

Sept 2025 — Apr 2026

- Co-developed a secure, production card-collection platform end-to-end using **Next.js**, **TypeScript**, **React**, and **Tailwind CSS**, deployed on Vercel and serving real users
- Designed and shipped scalable REST APIs and microservices using Next.js API routes, Prisma ORM, and PostgreSQL, owning the full backend lifecycle from schema design through live deployment and query optimization
- Built a stateful peer-to-peer trade-matching engine with complex relational queries across users, collections, and offers, applying data-structure and algorithm choices to keep response times low at scale
- Integrated a YOLO segmentation model behind a REST endpoint with embedding-based context and difference hashing, automating card identification in real time at **99% accuracy**
- Owned end-to-end feature delivery within a team of 7 — writing unit tests, running code reviews, and maintaining CI/CD pipelines with Git, GitHub Actions, and Docker

Personal Website (ishpreetnagi.com)

July 2024 — Present

- Built a responsive, single-page portfolio using **React**, **Astro**, and **Tailwind CSS**, with reusable components and accessible design patterns
- Optimized build pipeline and runtime performance, achieving a **95+ Lighthouse score** across performance, accessibility, best practices, and SEO

Technical Executive, DeltaHacks 10 (github.com/deltahacks/landing-10)

Aug 2023 — May 2024

- Shipped the official DeltaHacks 10 site to **1000+ applicants** under a hard deadline, working cross-functionally with design and operations leads
- Optimized build pipeline and bundle size using Astro and TypeScript, **increasing user engagement by 25%** through dynamic content and responsive design

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, SQL (PostgreSQL, MySQL), HTML/CSS
- **Frontend:** React, Next.js, Astro, Tailwind CSS, Chakra UI, Material UI, Redux (familiar), responsive/accessible UI
- **Backend & APIs:** Node.js, Next.js API routes, REST APIs, GraphQL (familiar), Prisma ORM, PostgreSQL, Redis (familiar), microservices, system design
- **Tooling & Delivery:** Git, GitHub Actions (CI/CD), Docker, Vercel, Supabase, automated testing, code review, TDD practices
- **AI in the Dev Loop:** GitHub Copilot, Cursor, Claude Code, OpenAI API, Claude API, LLM integration, RAG, prompt chaining